Ice Planed.

PERHAPS it will interest some of those who are investigating the structure of different kinds of ice to know how blocks of it may, with ease and certainty, be shaped into bars and plates of any required dimensions. Some time ago I had occasion to prepare some specimens for experimental purposes. At first my success in working the ice into the required form was not very great, for it cracked in all directions under the action of a saw or chisel. After trying many devices, I at last resorted to a joiner's plane; a tool which may have been tried for the purpose before, though I do not remember having seen its use suggested. With it ice may be planed with greater ease than wood, the shavings coming away in powder, and leaving the ice with a clean, bright, sound surface. R. M. DEELEY.

39 Caversham Road, Kentish Town, N.W., February 16.

REPETITION OF HERTZ'S EXPERIMENTS, AND DETERMINATION OF THE DIRECTION OF THE VIBRATION OF LIGHT.

SINCE last October, Prof. Fitzgerald and I have been repeating some of Prof. Hertz's experiments, as occasion allowed; and it may not be without interest at the present time to give a short account of our work.

The first experiment tried was the interference of direct electro-magnetic radiation with that reflected from a metallic sheet. This experiment is analogous to that known in optics as "Lloyd's experiment."

The radiation was produced by disturbances caused in the surrounding space by electrical oscillations in a conductor. It was arranged in this wise. Two thin brass plates, about 40 centimetres square, were suspended by silk threads at about 60 centimetres apart, so as to be in the same plane. Each plate carried a stiff wire furnished at the end with a brass knob. The knobs were about 3

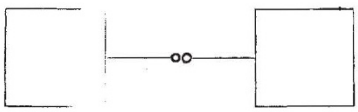


FIG. 1.

millimetres apart, so that on electrifying one plate a spark could easily pass to the other. This spark, as is well known, consists not simply of a transference of half the electricity of the first plate to the second—though this, which is the final state, is all that is observable by ordinary experimental methods—but the whole charge passes across to the second plate, then returns, and so on, pendulum-fashion, the moving part of the charge becoming less each time, till finally brought to rest, the energy set free at sparking being converted partly into heat in the wire and air break, partly into radiation into space, or in terms of action at a distance in inducing currents in other bodies.

The time taken by the charge to pass over to the second plate and to return, is a definite thing for a given sized arrangement, and depends on the connection between them. If C be the capacity of the plates, and I the self-induction of the connection, the time of each complete oscillation equals $2\pi\sqrt{CI}$. The time in the case of the particular arrangement used is, speaking roughly, about the 1/30,000,000 of a second.

If there be conductors in the neighbourhood of this "vibrator," currents will as usual be induced in each on every passage of the charge between the plates, each passage serving simply as a primary current.

Now, speaking briefly, the whole object of the experiment is to find out if these induced currents take place simultaneously in conductors situated at various distances from the primary current, and if not, to determine the delay. In order to do this we must, in the first place, be possessed of some means of even ascertaining that these currents occur, all ordinary methods being inadequate for detecting currents lasting only for such exceedingly short periods as these do. By devising how to determine the existence of these currents, Hertz made the experiment possible.

His method depends on the principle of resonance, previously suggested by Fitzgerald, and his current-observing apparatus is simply a conductor, generally a wire bent into an unclosed circle, which is of such a length that if a current be induced in it by a passage of a charge across the "vibrator" the return current or rush back of the electricity thus produced in the ends of the wire occurs simultaneously with the next impulse, due to the passage back across the "vibrator."

In this way the current in the "resonator" increases every time, so that at last the end charges, which are always of opposite sign, grow to be so great that sparks will actually occur if the ends of the wire are brought near together. Thus, Hertz surmounted the difficulty previously experienced by Fitzgerald when proposing electro-magnetic interference experiments.

The time of vibration in this circle is, as before, $2\pi\sqrt{CI}$, but on account of difficulties in calculating these quantities themselves, the length of the wire is most readily found by trial. To suit the "vibrator" we used, it was about 210 centimetres of wire No. 17. The ends of the wire were furnished with small brass knobs, which could be adjusted, as to distance between them, by a screw arrangement, the whole being mounted on a cross of wood for convenience in carrying about.

At first sight the simplest "resonator" to adopt would seem to be two more plates arranged similarly to the "vibrator," but it will be seen on consideration that it would not do, because no break for seeing the sparking could be put between the plates, for, if it were, the first induced current would be too feeble to jump the break, so that the reinforcement stage could never begin.[1]

The charging of the "vibrator" was effected by connecting the terminals of an induction-coil with the plates. In this way a continuous shower of sparks could be obtained in the resonating circle.

The circle in the interference experiment was held in the horizontal plane containing the axis of the "vibrator," the ends of the circle of wire being in such a position that a line joining the knobs was at right angles to the "vibrator." In this position only the magnetic part of

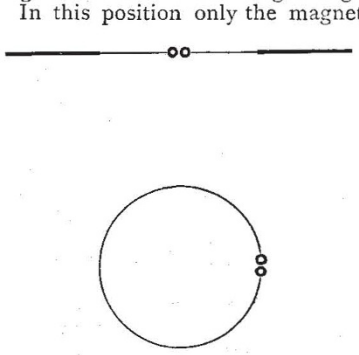


FIG. 2.

the disturbance could affect the circle, the "magnetic lines of force," which are concentric circles about the axis of the "vibrator," passing through the "resonator" circles.

When the knobs of the circle are brought round through 90°, so as to be parallel to the "vibrator," the electric part of the disturbance comes into play, the electric lines of force being, on the whole, parallel to the axis of the "vibrator." The electric action alone can cause a forced vibration in the knobs, even when the connecting wire is removed, if placed fairly close to the "vibrator."

[1] However, two pairs arranged in line, the pairs connected by a wire, could probably be got to spark between the centre plates.

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Again, if the knobs be kept in this position, but the circle be turned through 90°, so that its plane is vertical, only the electric part can act, the magnetic lines of force just grazing the circle. In this way the disturbance can be analyzed into its magnetic and electric constituents.

Lastly, if the knobs be in the first position, while the circle is vertical, there will be no action.

To exhibit these alone forms an interesting set of experiments. It also makes a very simple and beautiful experiments to take a wire twice as long and fix it instead of the first, but with two turns instead of one; no sparking is then found to occur. This is, of course, quite opposed to all ordinary notions, double the number of turns being always expected to give double the electromotive force. In this way the reality of the resonance is easily shown.

Interference Experiment.—The sparking, of course, becomes less intense as the resonator is carried away from the "vibrator," but by screwing the knobs nearer together it was possible to get sparks at 6 and 7 metres away. On bringing a large sheet of metal (3 metres square, consisting of sheet zinc) immediately behind the "resonator," when in sparking position, the sparking increased in brightness, and allowed the knobs to be taken further apart without the sparking ceasing; but when the sheet was placed at about 2·5 metres further back, the sparking ceased, and could not be obtained again by screwing up the knobs. On the other hand, when the sheet was placed at double this distance (about 5 metres), the sparking was slightly greater than without the sheet.

Now these *three* observations can only be explained by the interference and reinforcement of a direct action of the "vibrator" with one reflected from the metallic sheet, and in addition by the supposition that the action spreads out from the vibrator at a finite velocity. According to this explanation, in the first position the reflected part combines with the direct and reinforces its effects. In the second position—that of no sparking—the reflected effect, in going to the sheet and returning, has taken half the time of a complete vibration of the "vibrator," and so is in the phase opposite to the incident wave, and consequently interferes with it.

If it were possible to tell the direction of the current in a "resonator" at any moment, then, by employing two of them, and placing one just so much beyond the other that the currents induced in them were always in opposite directions, we would obtain directly the half-way length. Now, by reflection we virtually are put in possession of two "resonators," which we are enabled to place at this distance apart, although unable to tell more than whether there be a current or not.

The distance from the position of interference to the sheet is a quarter of the wave-length, being half the distance between these simultaneous positions of opposite effects.

In the third position the reflected wave meets the effect of the next current but one in the "vibrator" after the current it itself emanated from, and since these two currents are in the same direction their effects reinforce each other in the "resonator." This occurs at half the wave-length from the sheet.

The first two observations alone could be explained by action at a distance, by supposing the currents induced in the metallic sheet to oppose the direct action in the "resonator" everywhere, and by also supposing that, in the immediate neighbourhood of the sheet, the direct action is overmastered by that from the sheet, while at 2·5 metres away the two just neutralize each other.

On this explanation, at all distances further the direct action should be opposed by that from the sheet, so that the fact of being increased at 5 metres upsets this explanation. Again, behind the sheet, evidently on this supposition, the two actions should combine so as to increase the sparking, but instead of this the sparking was found to cease on placing the sheet in front of the "resonator."

In performing these experiments the "resonator" circle was always placed in the position in which only the magnetic part of the disturbance had effect. Hertz has also used the other positions of the resonating circle, whereby he has observed the existence of an electric disturbance coincident with the magnetic one, the two together forming the complete electro-magnetic wave.

Ordinary masonry walls were found to be transparent to radiation of this wave-length—that is, of about 10 metres—and some visitors to the opening meeting of the Dublin University Experimental Society, last November, were much astonished by seeing the sparking of the resonating circle out in the College Park, while the vibrator was in the laboratory.

Attempts were first made last December to obtain reflection from the surface of a non-conductor, with the hope of deciding by direct experiment whether the magnetic or electric disturbance was in the plane of polarization; that is, to find out whether the "axis of the vibrator" should be at right angles to the plane of reflection or in it, when at the polarizing angle, for obtaining a reflected radiation. It is to be observed that in these radiations the electric vibration is parallel to the "axis of the vibrator" while the magnetic is perpendicular to it, and that they are consequently polarized in the same sense as light is said to be polarized.

Two large glass doors were taken down and used for this purpose, but without success; and until lately, when reflection from a wall was tried, the experiment seemed unlikely to be successful.

In working with the glass plate, the resonator circle was first placed so that the "vibrator" had no effect on it. Then the glass plate was carried into position for reflection, but without result, though even the reflection from the attendants moving it was amply sufficient to be easily detected.

To obviate a difficulty arising from the fact that the wave was divergent, we decided to try Hertz's cylindrical parabolic mirrors, for concentrating the radiation. Two of these were made with sheets of zinc nailed to wooden frames, cut to the parabolic shape required.

In the "focal line" (which was made 12·5 centimetres from the vertex) of one of these, a "vibrator" was

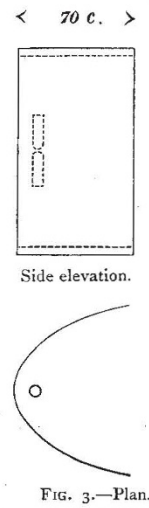


FIG. 3.—Plan.

placed, consisting of two brass cylinders in line, each about 12 centimetres long and 3 centimetres in diameter, rounded at the sparking ends.

In order that the "resonator" wire may lie in the "focal line" of the receiving mirror, it has to be straight;

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this necessitates having two of them. They each consist of a thick wire 50 centimetres long, lying in the "focal line," and of a thin wire, 15 centimetres long, attached to one end at right angles, and which passes out to the back of the mirror through a hole in the zinc, where the sparking can be viewed, without obstructing the radiation in front. The total length of each "resonator" is about two wave-lengths, the wave-length being about 33 centimetres, so that it may be that there are two vibrating segments in each of these "resonators."

With this apparatus it is possible to deal with definite angles of incidence. No effect was obtained with glass plates using these mirrors, whether the "vibrator" was perpendicular to the plane of reflection or in it. But with a wall 3 feet thick reflection was obtained, when the "vibrator" was perpendicular to the plane of reflection; but none, at least at the polarizing angle,[1] when turned through 90° so as to be in it.

This decides the point in question, the magnetic disturbance being found to be in the plane of polarization, the electric at right angles. Why the glass did not reflect was probably due to its thinness, the reflection from the front interfering with that from the back, this latter losing half a wave-length in reflection at a surface between a dense and a rare medium; and, as Mr. Joly pointed out, is in that case like the black spot in Newton's rings, or more exactly so, the black seen in very thin soap-bubbles. Hertz has pointed out several important things to be guarded against in making these

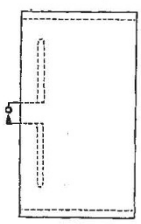


FIG. 4.

experiments. Ultra-violet light, for example, falling on the "vibrator," prevents it working properly, the sparking in the resonator ceasing or becoming poor. Also, the knobs of the "vibrator" must be cleaned, of burnt metal, and polished every quarter of an hour at least, to prevent a like result.

Both these effects probably arise, as suggested by Mr. Fitzgerald, from a sort of initial brush discharging (either ultra-violet light or points being capable of doing this), which prevents the discharging impulse being sufficiently sudden to start the oscillation in the "vibrator." For, to start a vibration, the time of impulse must be short compared with the time of oscillation. These precautions, therefore, become especially needful when working with small-sized "vibrators." Possibly, charging the "vibrator" very suddenly, after the manner of one of Dr. Lodge's anti-lightning-rod experiments, would save the irksome necessity of repeatedly cleaning the knobs of the "vibrator."

Several important problems seem to be quite within reach of solution by means of these Hertzian waves, such as dispersion. Thus, it could be tried whether placing between the reflector and the "resonator" conducting bodies of nearly the same period of vibration as the waves used would necessitate the position of the "resonator" being changed so as to retain complete interference. Or again, whether interspersing throughout the mass of a large Hertzian pitch-prism conductor with nearly the same period would alter the angle of refraction. In some such way as this, anomalous dispersion, with its particular case of ordinary dispersion, may yet be successfully imitated.

[1] Slight reflection was obtained at an incidence of 70°.

The determining the rate of propagation through a a large tile, or sheet of sandstone, could be easily made by means of the interference experiment, by placing it between the screen and the "resonator."

FRED. T. TROUTON.

THE SCHOOL OF FORESTRY AT DEHRA DOON, INDIA.

LAST year we gave an account of the newly-established School of Forestry at Cooper's Hill, the first of the kind in the United Kingdom, and explained what kind of instruction was there given, and how it was suited to the training of officers for the Indian Forest Department. We now propose to say something of its brother in India—an elder brother, indeed, by some eight years—the School at Dehra Doon, in the North-Western Provinces, now engaged in the education of those who may, not inaptly, be called the non-commissioned officers of the Department. The Dehra Doon is a long valley, which lies at the foot of that portion of the Himalaya which stretches between the great rivers Jumna and Ganges. It is shut off from the great Gangetic plain by a range of hills called the "Siwaliks," known well to all students of palæontological geology as the range in which were found the wonderful series of bones of extinct mammals described by Messrs. Falconer and Cautley. The valley itself lies about 2000 feet above the level of the sea, possesses a beautiful climate free from the blasts of the hot winds which, in April to June, sweep over the plains to the south of it; and is further known historically as having been the site of the first experiments made by the Indian Government in growing the tea-plant, experiments which proved its suitability to India, and made the Doon the fatherland of the great Indian tea industry—an industry which has gradually increased to such an extent that the exports of tea from India and Ceylon now very nearly rival in amount those from the Chinese Empire. Centrally situated in this beautiful valley, among plantations of tea, forests of sál-wood, and groves where the deodar of the Himalaya may be seen alongside of the mango, typical of the Indian plains, and feathery bamboos raise their heads from an undergrowth in which wild or semi-wild roses thrive with luxuriance, lies the town of Dehra Doon, the head-quarters of a Deputy-Commissioner, of the offices of the great Trigonometrical Survey of India, of a regiment of Ghoorka troops, and of the body-guard of the Viceroy. It is rather a straggling town, like most similar Indian stations; but, centrally situated and surrounded by gardens, is found the Forest School, of which we wish to convey some idea to our readers. The School was first started, in 1878, by the exertions of the then Inspector-General of Forests, now Sir Dietrich Brandis, K.C.I.E., and the first Director was Lieut.-Colonel F. Bailey, of the Royal Engineers.

At present the Director is Mr. W. R. Fisher, B.A. of Cambridge University, who is assisted by a Professor of Forestry, Mr. E. E. Fernandez, and a Professor of Geology and Chemistry, Dr. H. Warth. Mr. Fisher himself lectures on forest botany, while other officers, attached to the School for the management of the adjacent forests, teach mathematics, forest law, forest entomology, and surveying, the teaching of the last-named subject being especially fostered by the presence, in the same building, of the office of the Forest Survey, which is now engaged in the preparation of careful detailed maps of the great forest estate which Government possesses in India, and which bids fair to become, not only by its agricultural and climatic effects, but by its financial success, one of the most valuable of the revenue-yielding departments of the Empire.

Attached to the School is a well-equipped museum, containing a magnificent collection of accurately-named

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